



for ground fire, and the underpowered obsolescent aircraft usually thought suitable for reconnaissance were easy prey for enemy fighters. But, at the cost of heavy loss of life, comprehensive photomontages of trench systems were built up and used for selecting targets for the artillery.

When the heavy guns opened up, again the aircraft came into play. The gunners needed observers to tell them where the shells were landing so they could correct their range and direction. In decent weather, airborne observers could usually see where a shot was falling but there was no efficient way of communicating this to the ground until 1916, when some new aircraft were capable of carrying radio transmitters. Combining the use of radio with the “clock system” – a code

of number and letter coordinates that identified where a shell had fallen in relation to the target – created a reasonably efficient spotting technique. At the static battle of the Somme in 1916, the British found that no amount of aerial observation could make the artillery barrages actually work against troops in deep fortifications. In the mobile battles of 1918, however, thousands of airmen sacrificed their lives in an artillery-bombardment strategy that,

EINDECKER IN ACTION

This rare aerial photograph captures a Fokker Eindecker in action over France. The aircraft had a metal tubular frame that gave it strength in a high-speed dive, but in other ways it was not of an advanced design. It was one of the last important aircraft to use wing warping, rather than ailerons, for lateral control.



alone, would have failed, but employed with other arms and methods was very effective.

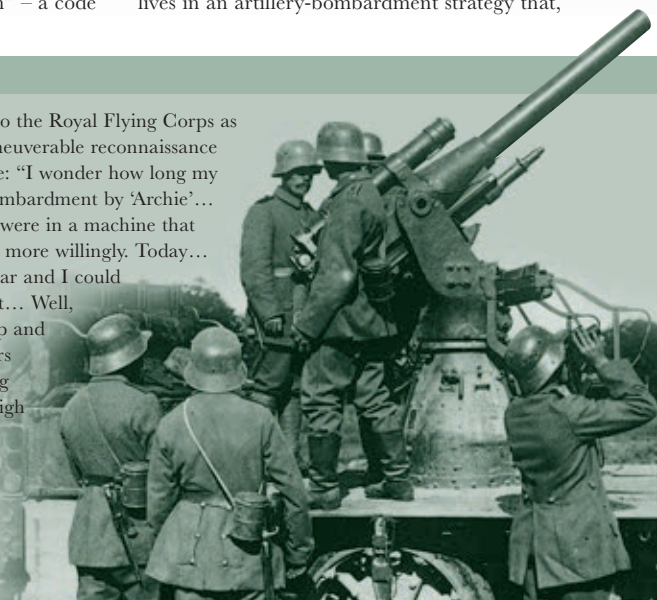
Toward the end of the war the Germans at last introduced a reconnaissance aircraft that gave its crew a reasonable measure of safety. The Rumpler C.VII could fly at 20,000ft (6,000m) and had an automatic camera that took a series of pictures when triggered. The downside was the lack of protection for high-altitude flight. Rumpler crews suffered from freezing cold (they were in open cockpits), and had to be supplied with oxygen.

FACING THE FLAK

GERMAN ANTI-AIRCRAFT FIRE, known to the Royal Flying Corps as “Archie,” took a heavy toll on unmaneuverable reconnaissance aircraft. Pilot Lt. William Read wrote: “I wonder how long my nerves will stand this almost daily bombardment by ‘Archie’... I would not mind quite so much if I were in a machine that was fast and that would climb a little more willingly. Today... some of the shells burst much too near and I could hear the pieces of shell whistling past... Well, I suppose the end will be pretty sharp and quick.” Most anti-aircraft commanders believed shrapnel was best for bringing down an aircraft. Others preferred high explosive or incendiary shells.

GOOD ODDS

German anti-aircraft gunners had a fair chance of hitting an observation aircraft if traveling at below 5,000ft (1,500m).



Arming the airplanes

For the generals, the main reason for putting guns in aircraft was to protect their own reconnaissance aircraft and shoot down the enemy's. But the initial impetus toward arming aircraft came from pilots and observers who simply wanted to “have a go” at the opposition. Firing pistols and carbines at passing aircraft had limited effect, while attempts at dropping grenades on them from above were a total failure. Machine guns were what were needed. But carrying such a weapon was a considerable burden for the lightweight, underpowered aircraft of 1914. It was also hazardous: there was a serious risk of blowing parts off your own